

MD. Mehedi Hasan Moon

Graduate Student · Department of Physics · Shahjalal University of Science and Technology

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RESEARCH PROFILE

Electronic structure theory applied to the relationship between molecular geometry and excited states, with the aim of designing organic emitters that can be built and measured rather than only predicted. Current work concerns π -stacked donor-acceptor triads in which a rigid U-shaped geometry forces through-space charge transfer, and the conditions under which the singlet-triplet gap becomes small enough for efficient reverse intersystem crossing. Routine user of Gaussian 16, ORCA 6.0 and Multiwfn; one first-author manuscript under review.

EDUCATION

Jun 2025 – ongoing	M.S. in Physics (Mixed Mode) Shahjalal University of Science and Technology, Sylhet Thesis: <i>Computational Design and Characterization of π-Stacked Donor-Acceptor Triads for TADF OLED Applications</i> . Supervisor: Prof. Dr. Nazia Chawdhury.
Jan 2019 – Apr 2025	B.Sc. (Honours) in Physics Shahjalal University of Science and Technology, Sylhet · CGPA 3.04 / 4.00 Project: <i>Numerical solution of a boundary value problem of the Legendre differential equation for P_4 using the finite difference method</i> . Supervisor: Prof. Dr. Mohammed Sujaul Haque Chowdhury.

MANUSCRIPTS

Under review	Bridge-Dependent Emission Tuning and Radiative-Rate Suppression in a Donor-Acceptor [3.3]Paracyclophane Series: A Gas-Phase TD-DFT Study MD. Mehedi Hasan Moon, Nazia Chawdhury – Department of Physics, SUST
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RESEARCH EXPERIENCE

11 Mar 2026 – present	Research Student · M.S. thesis Organic Optoelectronics Lab, Dept. of Physics, SUST – Prof. Dr. Nazia Chawdhury - Investigating intramolecular π-stacked donor-acceptor triads with rigid U-shaped geometries that enable through-space charge transfer. - Studying thermally activated delayed fluorescence as a route to the efficiency roll-off problem in OLED devices. - Ground-state configuration and geometry optimisation in Gaussian 16 with dispersion-corrected functionals. - Hole-electron distribution analysis in Multiwfn to characterise electronic transitions. - Optimising molecular architecture to hold ΔE_{ST} below 0.1 eV for efficient reverse intersystem crossing.
11 May 2025 – 11 Mar 2026	Research Assistant Department of Physics, SUST Computational work on organic emitters — geometry optimisation, excited-state calculations and wavefunction analysis supporting the group's TADF programme.
2025 – ongoing	Time-Dependent Neutrino Emission from Neutron-Capture Processes: A Framework for Coupling SkyNet with MESA Department of Physics, SUST – Anock Somadder, Associate Professor Coupling a nuclear reaction network to a stellar evolution code so that neutrino emission can be followed as a time-resolved output rather than estimated afterwards.

Jun 2023 –
Dec 2023

Undergraduate project

Department of Physics, SUST – Prof. Dr. Mohammed Sujaul Haque Chowdhury

Numerical solution of a boundary value problem of the Legendre differential equation for P_4 by the finite difference method.

RESEARCH INTERESTS

Primary	Organic optoelectronics, through-space charge transfer, TADF emitters.
Secondary	Organic photovoltaics, charge-carrier mobility in π -stacked systems, exciton dissociation and transport, donor-acceptor bulk heterojunctions.
Technical	Computational materials science, density functional theory, molecular structure-property relationships.
Broader	Organic semiconductors, π -stacked functional materials, solar energy, energy storage, heat transfer, nonlinear optics.

TECHNICAL SKILLS

Computational software	ORCA 6.0 / Gaussian 16 / Multiwfn
Theoretical frameworks	Density functional theory / TD-DFT / PCM solvation
Programming	Python / C++ / Bash and shell scripting / Mathematica
Libraries	Matplotlib
Environment	Linux / Git & GitHub / Zotero / Word, Excel, PowerPoint
Languages	Bengali – native / English

TRAINING

7 Jan 2026	Research Workshop: The Path to Academic Excellence SUST Research Society (SUSTRS) – participant Research ethics and integrity, strategies for publishing in Q1/Q2 journals, structuring research papers, and citation management with Zotero.
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TEACHING

Supplementary instruction	Supplementary Instructor • physics and mathematics Secondary and post-secondary students Designed and delivered lesson plans built around experiential learning and practical application, and assessed coursework with feedback aimed at measurable improvement.
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INSTITUTIONAL SERVICE

18 Dec 2025	Convener International Conference for Advances in Physics, Dept. of Physics, SUST Led the organising committee across logistics, scheduling and technical sessions for international participants, working with department faculty to deliver the conference's academic programme.
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HONOURS AND AWARDS

2015	Government Scholarship Awarded on Secondary School Certificate results
2012	Government Scholarship Awarded on Junior School Certificate results

References available on request.